

Reading and Writing Module 1

27 QUESTIONS

1

Blue holes, such as Dean's Blue Hole and the Tar Side Blue Hole, are deep sinkholes that form underwater. Because the dark blue appearance of their waters means these pits can be easily_____shallow areas around them, where the water looks light blue, many blue holes have been discovered by people doing recreational activities like boating.

Which choice completes the text with the most logical and precise word or phrase?

- A) surpassed by
- B) elevated from
- C) differentiated from
- D) explored by

2

Although the cumulative weight of the goods in a cargo truck affects fuel efficiency, this effect --- and the corresponding effect on the cost of a given trip to the trucking company --- is relatively small. The large majority of a trip's cost stems from the mere fact of the truck being on the road; accordingly, trucking firms are_____to dispatch trucks that aren't full.

Which choice completes the text with the most logical and precise word or phrase?

- A) admonished
- B) entitled
- C) loath
- D) prone

3

By eliminating middlemen, direct-to-consumer sales channels allow small farms to capture retail prices rather than accept lower wholesale rates. The pricing advantages that come with adopting this model, combined with the customer loyalty that comes from forging personal connections, can_____the financial outlook for many small agricultural producers.

Which choice completes the text with the most logical and precise word or phrase?

- A) transform
- B) hasten
- C) overshadow
- D) interpret

4

Sazami (Ripples) is an important work of Nihonga, or classical Japanese painting. Unlike many Artists who adopted traditional European methods such as painting with oil on canvas, Fukuda Heihachiro_____ traditional Japanese approaches. For instance, Fukuda produced Sazami by applying color and platinum to a silk surface.

Which choice completes the text with the most logical and precise word or phrase?

- A) adhered to
- B) awarded
- C) reprimanded
- D) refrained from

5

Before refrigeration, food was preserved using salt, smoking, or ice harvested from frozen lakes. The invention of the electric refrigerator changed how some people stored and preserved their food. In some countries, this new technology became more commonly available in homes during the 1920s and 1930s. People in these places were now able to keep food cold and fresh for extended periods. As a result, they were able to shop for certain foods and drinks less frequently.

Which choice best states the main purpose of the text?

- A) To describe the health benefits of storing food at the correct temperatures over long periods of time
- B) To discuss how the invention of the refrigerator changed people's food storage and shopping habits
- C) To explain the science behind keeping different foods preserved with and without refrigeration
- D) To convince readers that modern conveniences such as refrigeration have led to a loss of critical skills

6

Human eyes contain cone cells that help process color and bright light. There are three types of cones: long, medium, and short. In a study, James Fong and his team used a laser technology to activate only the medium cones in participants' eyes. When they did this, the participants saw a new blue-green color that the researchers then named olio.

Which choice best describes the overall structure of the text?

- A) It discusses some characteristics of cones in human eyes and then discusses a study involving those cones.
- B) It mentions three different colors and then mentions the wavelengths associated with those colors.
- C) It names different cells in human eyes and then states how they work with cells in a different part of the body.
- D) It explains what the color olio is and then explains how that color was discovered.

7

Text 1

In parts of New Zealand, the stoat is a major predator of the house mouse. Researcher G. L. Blackwell and colleagues found that when this predation pressure on house mice was temporarily reduced, their numbers significantly increased. This finding illustrates a foundational ecological principle:

predators control prey population numbers.

Text 2

François Potvin and colleagues found that excluding wolves from a site in Quebec, Canada, where they typically prey on white-tailed deer had no significant effect on white-tailed deer abundance. Many other predation-reduction studies show an increase in prey abundance, but those studies often focus on small, rapidly reproducing prey, like birds, mice, and frogs, rather than large, slowly reproducing prey, like white-tailed deer, which could account for the difference between these results and Potvin and colleagues' results.

Based on the texts, the author of Text 2 would most likely agree with which statement about the "principle" mentioned in Text 1?

- A) It has been challenged by some studies, but the findings of those studies have not been widely accepted.
- B) It is plausible, but many of the studies that support it have methodological flaws.
- C) It has some evidential support, but it should not be regarded as universally applicable.
- D) It may be true for some predators but only because those predators share certain physical characteristics.

8

While not everyone notices it, beneath the tree canopy of an ordinary city park lies an extraordinary web of life --- fungus breaking down fallen leaves, native bees pollinating wildflowers, and moss growing in the crevices of an old log. To document this hidden biodiversity, everyday people have been participating in "bioblitz" events, in which participants photograph, collect, and identify species using both field guides and mobile apps. These collective observations reveal which plant and animal species depend on one another for survival --- ecological relationships that might otherwise remain unseen.

In context, the text's characterization of the biodiversity in city parks as "hidden" most clearly conveys which idea?

- A) The environmental interactions in city parks are abundant but occur primarily during nighttime hours when human visitors are absent.
- B) Some wildlife species that populate city parks have developed sophisticated camouflage techniques that minimize their visibility.
- C) The ecological relationships in city parks may not be readily apparent without systematic observation and documentation efforts.
- D) There are many beautiful parks in cities, but they often remain unseen because of poor urban planning decisions.

9

In environments ranging from shallow tidal areas to the deep ocean, single-celled marine foraminifera have been found to contain phosphate at concentrations many times greater than those in the surrounding water. Notably, though, species from oxygen-rich environments like the Mid-Atlantic Ridge show lower phosphate concentrations than species from oxygen-poor areas. Scientists have thus suggested that especially high internal phosphate concentrations enable foraminifera to offset the effects of low oxygen.

Which choice best states the main idea of the text?

- A) There is a correlation between habitat depth and phosphate concentration in marine foraminifera, with deeper-dwelling species generally maintaining higher internal concentrations.
- B) While marine foraminifera in general have high phosphate concentrations, those concentrations tend to vary by species' habitat, possibly reflecting adaptations to different levels of oxygen.
- C) Marine foraminifera maintain similar phosphate concentrations across different ocean environments, suggesting that while this trait plays a role in offsetting low oxygen levels, that is not its only function.
- D) Marine foraminifera counterbalance the effects of reduced oxygen levels by storing phosphate at high concentrations, matching or in some cases even slightly exceeding the phosphate concentration in the surrounding seawater.

10

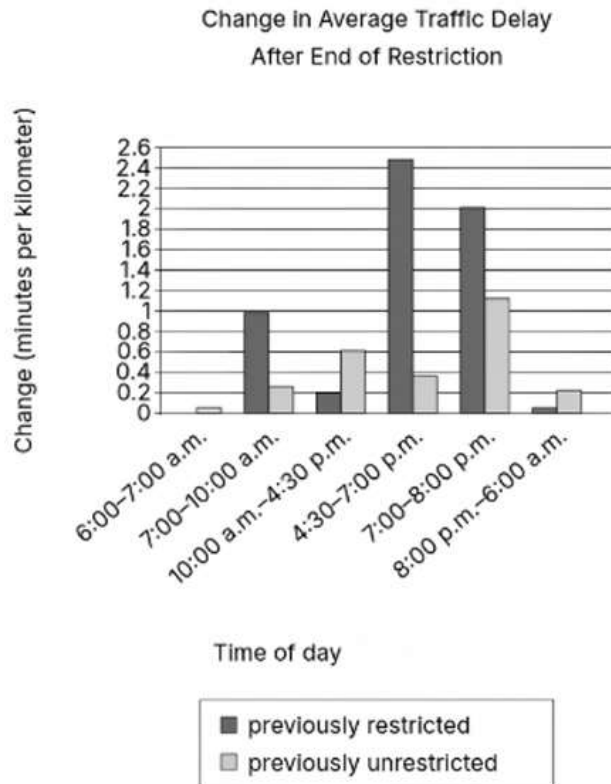
Film name	Visible light Allowed through	Glare reduction
No film used	90%	14%
GlowGuard	33%	52%
ShieldView	22%	65%
BronzeShine	16%	77%
ShadeBlock	7%	82%

A restaurant is installing window film to create a more comfortable indoor atmosphere. This is because during dinner service, the sunlight gets in people's eyes while they eat. The restaurant owner wants to reduce the glare but still let customers enjoy the sunset. To help meet these goals, a window-film technician recommends selecting a film that reduces glare as much as possible while still allowing more than 15% of visible light to pass through the windows. The table shows the performance data for several different films.

Based on the window-film technician's recommendation and the data in the table, which film should the restaurant owner choose?

- A) BronzeShine
- B) ShieldView
- C) ShadeBlock
- D) GlowGuard

11



In 2016, the city of Jakarta --- one of the world's largest metropolitan areas -- revoked a rule that restricted the use of major roads in the city's business district to only vehicles with three or more passengers during rush hours (7:00-10:00 a. m. and 4:30-7:00 p. m.). As researchers had expected, this move led to increased traffic delays during those times on the roads whose use had been limited, especially in the late rush hour period. Surprisingly, delays also worsened during other periods of time and on roads that hadn't been subject to the rule; data most clearly reflect this effect in the period of _____

Which choice most effectively uses data from the graph to complete the statement?

- A) 10:00 a. m.-4:30 p. m. for previously unrestricted roads.
- B) 7:00-10:00 a. m. for both types of roads.
- C) 7:00-8:00 p. m. for both types of roads.
- D) 4:30-7:30 p. m. for previously restricted roads.

12

A film studies student is researching early 20th-century film serials, which consisted of individual episodes of a single long story that were shown weekly in theaters. The Shadow of the Eagle is a 1932 serial that, over its 12 episodes, kept its audience interested with the suspense and drama that are typical of the crime genre. The student, however, claims that ultimately audiences of the time preferred resolution and closure over ongoing tension.

Which finding, if true, would most directly support the student's claim?

- A) Audiences of the time considered The Shadow of the Eagle to belong to a genre other than the crime genre.
- B) The 12th episode of The Shadow of the Eagle was the most expensive episode of the series to produce.
- C) Modern critics generally regard the first episode as the best installment of The Shadow of the Eagle.

D) The 12th episode of *The Shadow of the Eagle* was viewed by more people than was any previous episode in the series.

13

Cane is a 1923 novel by Jean Toomer. In the novel, Toomer mentions a road in rural Georgia called Dixie Pike and describes it as having a deep connection to a faraway place, writing, _____

Which quotation from *Cane* most effectively illustrates the claim?

- A) "And when the wind is from the South, soil of my homeland falls like a fertile shower upon the lean streets of [Washington, DC]."
- B) "The Dixie Pike has grown from a goat path in Africa."
- C) "If it were dusk, then [Fern's eyes would] wait for the search-light of the evening train which you could see miles up the track before it flared across the Dixie Pike, close to her home."
- D) "I leave the men around the stove to follow [Carma] with my eyes down the red dust road."

14

The advent of online streaming has led many music listeners to drift away from ownership of music (through downloads or through physical media such as compact discs) and toward the streaming services iHeartRadio and Tidal, among others. Data et al. studied the impact of this change on the variety of music that listeners consume. The researchers reasoned that the ownership model of music assigns a cost per song to acquiring a variety of music, while streaming services typically charge a flat fee for access to an entire music catalog, making variety free, which suggests that _____

Which choice most logically completes the text?

- A) listeners who use streaming services would be more likely to give physical copies of music as gifts to others than to purchase physical copies for themselves.
- B) music publishers who choose to forgo releasing music on physical media are likely to see no change in revenue.
- C) the music choices of listeners who use streaming services would likely be more varied than those of listeners who do not use streaming services.
- D) listeners who prefer to purchase compact discs rather than use a service such as iHeartRadio or Tidal would tend to listen to older music.

15

It is hard for publishers and printers to create very long books that balance the requirements of readability, durability, and cost. Among some exceptionally long novels published in English, Miguel de Cervantes' *Don Quixote* (translated from Spanish) was published in a 1,239-page edition and runs to an estimated 590,000 words, while Margaret Young' s *Miss Macintosh, My Darling* (written originally in English) was published in a 1,198-page edition and runs to an estimated 750,000 words. This suggests that_____.

Which choice most logically completes the text?

- A) readers will, on average, take the same time to read *Miss Macintosh, My Darling* as they will to read *Don Quixote*.
- B) the edition of *Don Quixote* was less expensive to print than the edition of *Miss Macintosh, My Darling*.
- C) the edition of *Miss Macintosh, My Darling* has more words per page than the edition of *Don Quixote*.

D) Don Quixote likely has a lower word count in its original Spanish edition.

16

On entering the produce section of the grocery store, the performer _____ not just the general scent of fruit but the distinct olfactory nuances of lemon (*Citrus limon*), camu camu (*Myrciaria dubia*), and hawthorn (*Crataegus rhipidophylla*).

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) distinguished
- B) distinguishing
- C) distinguish
- D) to distinguish

17

An accomplished birder might be able to spot thousands of different species over the course of a career, from *Treron sphenurus* to *Lanius ludovicianus*, but no birder has ever achieved the ultimate _____ an in-person glimpse of all 10,000-plus species of birds.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) goal: catching
- B) goal, catching
- C) goal; catching
- D) goal. Catching

18

In their 2019 study of butterfly diversity at Rupa Lake in Nepal, researcher Bijaya Neupane and his colleagues counted one blue-lail jester _____ to the Nymphalidae family, the blue-tail jester (*Symbrenthia niphanda*) is known to be very rare.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) butterfly and belonging
- B) butterfly. Belonging
- C) butterfly belonging
- D) butterfly, belonging

19

With evidence of long-term use over centuries (a pattern rarely seen in early Caribbean sites) _____ conventional theories about highly mobile foraging groups, the Ortiz site is an example of what archaeologists call persistent placemaking.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) have challenged
- B) has challenged
- C) challenging
- D) challenged

20

To experience both the multi-instrument groove of contemporary blues music and the stripped-down Delta blues of Houston Stackhouse and other blues forebears ____ the rich history of the genre.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) is to realize
- B) are realizing
- C) realizing
- D) to realize

21

When historian John Irwin founded the Museum of Appalachia in Clinton, Tennessee, in the 1960s, it consisted of just a single log structure. Today, it is a 65 acre living history museum featuring vast collections of folk art, including carvings, textiles, _____ the Arwine Cabin, one of thirty five historic log buildings on the property; and a recreated pioneer era Appalachian farm.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) paintings,
- B) paintings;
- C) paintings
- D) paintings.

22

The Dave Dashaway book series was among the many children's series that were published during the first half of the twentieth century. ____ the Dave Dashaway books were published from 1913 to 1915.

Which choice completes the text with the most logical transition?

- A) To be exact,
- B) Finally,
- C) Nevertheless,
- D) By contrast,

23

The ancient Inca built a massive network of roads connecting all parts of their mountain empire. _____ messengers could quickly travel to even the most distant villages in just a few days.

Which choice completes the text with the most logical transition?

- A) As a result,
- B) Tomorrow,
- C) Last time,
- D) Besides,

24

Meteorologists use various specialized tools to measure atmospheric conditions. _____ barometers detect changes in air pressure that often signal approaching weather systems, while anemometers measure wind speed and direction.

Which choice completes the text with the most logical transition?

- A) For instance,
- B) Similarly,
- C) Next,
- D) As a result,

25

As anthropologist Cristina Grasseni explains, many artisanal cheeses in Italy, such as Robiola di Roccaverano from Piedmont and Casatella Trevigiana from Veneto, are highly valued for their regional authenticity: _____ they are prized for being made using local ingredients and methods typical of their respective geographic regions.

Which choice completes the text with the most logical transition?

- A) that is,
- B) then,
- C) rather,
- D) still,

26

- India Arie is an African American singer and songwriter.
- The media outlet All Music has described her music as having "glistening acoustic guitar, churchy organ, and smooth, supple beats."
- Her first studio album, *Acoustic Soul*, was released in 2001.
- It features the song "Always in My Head."
- Ralph Rolle played drums on the album.

Which choice most effectively uses information from the given sentences to introduce *Acoustic Soul* to an audience already familiar with India Arie?

- A) India Arie's music has been described as having "glistening acoustic guitar, churchy organ, and smooth, supple beats."
- B) Featuring the song "Always in My Head," *Acoustic Soul* is India Arie's first studio album.
- C) Ralph Rolle played drums on singer and songwriter India Arie's 2001 album, *Acoustic Soul*.
- D) "Always in My Head," a song by singer and songwriter India Arie, is featured on the album *Acoustic Soul*.

27

- The Highpointers Club is a hiking club.
- One of the main goals among club members is to reach the highest points in all fifty US states.
- Those who achieve this are called 50 Completers.
- In Suk Han became a 50 Completer on November 15, 2013.

- The highest point in Michigan is Mount Arvon, at 1,979 ft.
- The highest point in New Mexico is Wheeler Peak, at 13,167 ft.

Which choice most effectively uses information from the given sentences to explain the 50 Completers hiking challenge to a new audience?

- A) Hikers aiming to count themselves among the 50 Completers must reach not only Michigan's Mount Arvon but also the even higher peak of Wheeler Peak in New Mexico.
- B) Not until after you have reached the highest points in all fifty US states—including Mount Arvon in Michigan and Wheeler Peak in New Mexico—can you include yourself among the 50 Completers of the Highpointers Club.
- C) If you are looking for a new hiking challenge, consider joining the Highpointers Club, as did In Suk Han, a hiker who successfully reached the highest point in every US state.
- D) On November 15, 2013, In Suk Han finally completed the feat of reaching the highest point in all fifty US states, including Spruce Knob in West Virginia and Mount Elbert in Colorado.

Reading and Writing Module 2

27 QUESTIONS

1

Though research indicating that plants may preferentially absorb copper and other essential elements (i. e., those necessary for plant growth) from soil over nonessential ones seems to suggest that copper-contaminated soil may be especially _____ to decontamination by plants, Milena Mohri and team found that plants tended to accumulate the nonessential element cadmium at higher rates than copper; the presence of cadmium in such soils should thus be assessed before pursuing plant-based copper decontamination.

Which choice completes the text with the most logical and precise word or phrase?

- A) pertinent
- B) detrimental
- C) unresponsive
- D) amenable

2

Despite the marked divergence of such ballads as "The House Carpenter" from their antecedents in Britain, the balladry of the Appalachian South long figured as a proxy for the music naturalized there in the colonial era. More recent scholarship offers a more _____ directly acknowledging Appalachian innovation to British ballads as well as the influence of spirituals sung in Black churches in the South.

Which choice completes the text with the most logical and precise word or phrase?

- A) recapitulation
- B) corollary

- C) corrective
- D) hypothesis

3

The presumptive functionality of khipu, the semiotic system of knotted cords once used by the Inka, to encode language (perhaps, like ancient Chinese script, in units of discrete words) now has an empirical basis: Manuel Medrano's statistical data on Spanish translations of khipu content, as recited by Quechua-speaking scribes, _____ historical reports of khipu's status as a linguistic record.

Which choice completes the text with the most logical and precise word or phrase?

- A) replicate
- B) complicate
- C) supersede
- D) supplement

4

For decades, automotive technicians focused almost exclusively on mechanical systems. However, the rapid adoption of electric vehicles has _____ this approach, as technicians must now master high-voltage safety protocols, battery management systems, and specialized diagnostic procedures that were entirely unnecessary in the era of internal combustion engines.

Which choice completes the text with the most logical and precise word or phrase?

- A) upended
- B) standardized
- C) misrepresented
- D) preceded

5

A variety of technologies make use of magnets incorporating rare earth elements, such as gadolinium. For technology manufacturers, dependence on rare earth elements is a significant _____: such elements are extracted at large scale in only a small number of locations, making any supply chain requiring them highly fragile.

Which choice completes the text with the most logical and precise word or phrase?

- A) redundancy
- B) discrepancy
- C) vulnerability
- D) insolvency

6

Marketing research recognizes three distinct types of price endings: round (like 25.00), just-below (like 24.99), and precise (like \$24.63). A number of studies have documented significant differences in how frequently these price endings appear in different countries' marketplaces. Although these variations could be attributable to cultural factors, researchers have yet to thoroughly investigate that connection, and it remains unclear which cultural differences may be relevant or how they might interact with other socioeconomic conditions.

Which choice best states the main purpose of the text?

- A) To analyze consumer purchasing behavior across different cultures
- B) To contrast cultural and socioeconomic explanations for differences in price endings
- C) To present the finding that three different pricing strategies are used broadly across cultures
- D) To identify a research gap regarding potential cultural influences on price endings

7

A light-year measures distance, not time, despite its potentially misleading name. It represents the distance that light travels through a vacuum in one Earth year — approximately 5.88 trillion miles or 9.46 trillion kilometers. Astronomers use this unit because conventional distance measures are impractical for cosmic scales. For context, Proxima Centauri, the nearest star to our solar system, is 4.24 light-years away, while the Andromeda Galaxy lies approximately 2.537 million light-years from Earth.

Which choice best describes the overall structure of the text?

- A) It highlights fundamental challenges in measuring vast cosmic distances, identifies limitations of conventional measurement units, proposes alternative approaches to conceptualizing space, and suggests how astronomical measurements might be improved in the future.
- B) It corrects a possible misconception about a scientific term, provides its precise definition with numerical equivalents, explains why the unit is used in astronomy, and offers examples of its application to cosmic objects.
- C) It contrasts various astronomical measurement systems, analyzes the advantages and limitations of each method, evaluates their accuracy for different cosmic scales, and demonstrates how they complement each other.
- D) It chronicles the historical evolution of astronomical measurement techniques, describes how earlier distance calculations were performed, explains how the light-year concept was developed by scientists, and gives two examples of distances measured in light-years.

8

Text 1

For decades, ornithologists assumed that if they saw a singing Bell's vireo — a bird species found in temperate North America — they must be observing a male trying to attract a mate or claim territory. As Peter J. B. Slater and Nigel I. Mann have emphasized, however, a similar assumption can't be made about birds in the tropics, where females sing as often as males do. Slater and Mann call for more research on this discrepancy between tropical and temperate female birdsong.

Text 2

Recent evidence shows that a female Bell's vireo is as capable of song as a male is. In fact, female birdsong is much more common among temperate species than currently assumed, claim Evangeline Rose and colleagues. These female songbirds sing less frequently than males do, and in duller tones, making it "easy for researchers to miss the quiet and hidden females and focus on the loud and colorful males," says Rose.

Based on the texts, how would Rose and colleagues (Text 2) most likely respond to the assertion by Slater and Mann (Text 1) about the different prevalence of female birdsong in temperate and tropical areas?

- A) They would raise the possibility that the difference in prevalence may be due to differences in the timing of the mating season among temperate and tropical bird species.

- B) They would argue that the apparent difference in prevalence may partly reflect a difference in the ease with which female birdsong and male birdsong can be detected.
- C) They would caution that the seeming difference in prevalence may be an artifact of researchers' tendency to study birdsong among temperate species more frequently than among tropical species.
- D) They would concede that the geographic difference in prevalence is real but argue that the frequency with which male tropical birds sing has been overstated by previous researchers.

9

In 2005, Swedish researchers studying flawed choice recall devised an experiment in which participants were shown two photos of different faces and asked to select the one they found more attractive. The researchers appeared to give the participants the photo they had chosen, but in fact used sleight-of-hand to give them the photo of the person they had not chosen. Most participants accepted the swap and, when asked to explain their initial preference, gave justifications for having chosen the face in the photo they were given. Subsequent researchers have replicated this finding with other choices, including jam preferences.

Based on the text, why was the sleight-of-hand step an important part of the experiment?

- A) It helped the researchers test whether participants would embrace an outcome they thought was consistent with their intention even when it was truly contradictory.
- B) It enabled the researchers to determine whether the thought process involved in selecting which photo had the more attractive face was similar to the thought process involved in selecting a preferred jam.
- C) It allowed the researchers to control for the possibility that some participants found the two faces roughly equally attractive.
- D) It allowed the researchers to measure the willingness of participants to change their positions when presented with persuasive counterarguments.

10

The following text is from George Eliot's 1860 novel *The Mill on the Floss*. The narrator is describing the fictional town of St. Ogg's, located in a part of Great Britain that was occupied by Romans and subsequently by Vikings in the ancient past.

It is one of those old, old towns which impress one as a continuation and outgrowth of nature, as much as the nests of the bower-birds or the winding galleries of the white ants; a town which carries the traces of its long growth and history like a millennial tree, and has sprung up and developed in the same spot between the river and the low hill from the time when the Roman legions turned their backs on it from the camp on the hillside, and the long-haired sea-kings came up the river and looked with fierce, eager eyes at the fatness of the land.

Which choice best captures St. Ogg's as it is presented by the narrator in the text?

- A) Because of its strategic location and abundant resources, St. Ogg's bears the marks of innumerable conflicts between groups who have vied for control over the town.
- B) By virtue of its great age, St. Ogg's has acquired the character of an organic entity rooted in its natural surroundings and marked by various external influences.
- C) In the course of its varied history, St. Ogg's has undergone such extensive transformations that its current incarnation would scarcely be recognizable to its earliest settlers.

D) As a result of its extreme antiquity, St. Ogg's tends to strike observers as dilapidated and haphazardly constructed.

11

Mineral	formation stage	$^{87}\text{Sr}/^{88}\text{Sr}$
Crystal	core	0.71685
Crystal	rim	0.71548
Crystal	matrix	0.71416

A student is studying igneous rock formations on Elba Island, Italy, and reviews published data on the ratio of strontium isotopes ($^{87}\text{Sr}/^{88}\text{Sr}$) in minerals from a specific rock sample. These isotope ratios provide information about the composition of the magma that formed the rock. Ratios below 0.710 suggest more material from the mantle in the magma, while ratios increasing above that number usually indicate greater proportions of material from the continental crust. The minerals in the rock sample record three growth stages: crystal core zones formed first, then rim zones, and finally the surrounding matrix. The student concludes that the rock sample_____

Based on the information given, which choice most effectively uses data from the table to complete the student's conclusion?

- A) does not show the expected $^{87}\text{Sr}/^{88}\text{Sr}$ ratios across the stages of mineral formation.
- B) contains $^{87}\text{Sr}/^{88}\text{Sr}$ ratios too similar to provide reliable information about magma sources.
- C) formed from magma with decreasing but persistent crustal influence.
- D) shows evidence of increasing continental crust material in later forming minerals.

12

A student is writing a research paper on the history of irrigation in the southwestern United States, situating the development of the Salt Spring Valley Reservoir (a man made body of water created in Calaveras County, California, in 1882) in a historical context. The student claims that some modern day irrigation methods used in southwestern states, such as California, Arizona, and New Mexico, have a lineage that predates the completion of the Salt Spring Valley Reservoir.

What quotation from a historian best supports the student's claim?

- A) "In the decades following the Second World War, irrigation of arid areas for agriculture increased to such a degree that it now accounts for roughly 70% of the world's water usage."
- B) "The importance of the development of irrigation infrastructure in the American Southwest today cannot be overstated, since it is the most common means of conveying water for food production."
- C) "The irrigation system developed by the Hohokam people in what is now central Arizona in the 7th century CE was simple but made use of hydraulic engineering design principles that are still utilized by today's engineers."
- D) "Sprinkler irrigation systems, which were developed in the late 20th century, are a convenient technique of irrigating that makes use of machinery in order to spray water in all directions. Many areas of the Southwestern United States are currently irrigated by this method."

13

A student is writing an essay about how writers of the late 1700s responded to the French Revolution and wants to include two pieces of textual evidence that show closely aligned attitudes toward the revolution. The student has selected the following quotation, adapted from Helen Maria Williams's

1791 poem "A Farewell, for Two Years, to England":

In new-born majesty [France] seems to rise,

While sudden from the land oppression flies.

So, at the solemn hour of Nature's birth,

When brooding darkness veil'd the beautiful earth,

Heaven's awful mandate pierc'd the solid night,

"Let there be light," it said, "and there was light!"

Which quotation from another prominent writer of the same period would be most effective for the student to include along with the quotation from Williams's poem?

- A) "The age of chivalry is gone. That of sophisters, economists, and calculators has succeeded; and the glory of Europe is extinguished forever." (Edmund Burke)
- B) "Let human nature, conscious of its own infirmities, humbly solicit the protection of Omnipotence against the arrogance of human Reason, whenever, proud of her own supremacy, she presumes to pass the interdicted bounds prescribed to her finite powers." (Jane West)
- C) "[Liberty] scorns to disdain / Her righteous cause with carnage, and resigns / To Fraud and Anarchy the infuriate crowd." (Charlotte Smith)
- D) "Reason has, at last, shown her captivating face, beaming with benevolence; and it will be impossible for the dark hand of despotism again to obscure its radiance." (Mary Wollstonecraft)

14

In 1967, archaeologist James Mellaart claimed that at its peak, Çatalhöyük --- a 33-acre Neolithic village in present-day Turkey that dates from approximately 7100 to 5950 BCE --- housed between 8,000 to 10,000 residents, an estimate based on the excavation of dense clusters of mud brick buildings that were thought to have been dwellings. Arguing that such estimates should be reevaluated, Ian Kuijt and Arkadiusz Marciniak (2024) cite radiocarbon data and ethnographic modeling that suggest the heterogeneity of building use in Çatalhöyük, with many individual buildings being inhabited for relatively brief periods --- 20 to 30 years or less --- before being abandoned or repurposed for nonresidential functions (e.g., storage). Accordingly, these recent findings challenge the assumption that _____.

Which choice most logically completes the text?

- A) the buildings excavated at Çatalhöyük at the time of Mellaart's research had served a residential purpose as early as 7100 BCE.
- B) the density of excavated buildings at Çatalhöyük can serve as a proxy for how many people inhabited the village at a given time.
- C) the use of mud bricks is a reliable indicator that a structure had been used either as a dwelling or for nonresidential purposes.
- D) the number of buildings at Çatalhöyük would have been sufficient to accommodate a population that expanded from 8,000 to 10,000 inhabitants by 5950 BCE.

15

In 2014, Marianne Dehasque et al. challenged earlier theories on causes of woolly mammoth extinction that emphasized isolation and inbreeding and a resulting decrease in genetic variation and increase in chances of harmful genetic mutations. Focusing on Wrangel Island off the coast of Siberia, where the last mammoths lived, the team used genome-based simulations to evaluate mutations in the population there. The analysis showed that particularly damaging mutations became less frequent, while moderate ones accumulated over time within the isolated population, which quickly grew from about 8 founding individuals to the 200-300 individuals it maintained until its extinction. The team thus concluded that despite its isolation, the Wrangel Island mammoth population likely _____.

Which choice most logically completes the text?

- A) developed increasing resistance to certain diseases and environmental harms over time.
- B) didn't succumb as a result of the most harmful genetic mutations becoming too prevalent for the population to be sustained.
- C) reached a size similar to that of nearby mammoth populations that were not isolated and thus had fewer genetic mutations.
- D) didn't go extinct suddenly but rather experienced a slow decline over its last several generations.

16

Walter De Maria's Earth Room installations in Germany and New York are notable examples of interior earth _____ former appearing in Munich and Darmstadt as temporary exhibits during the 1970s and the latter becoming a permanent fixture in Manhattan in the 1980s. The New York Earth Room consists of 280,000 pounds of soil spread across 3,600 square feet of gallery space.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) sculpture, the
- B) sculpture
- C) sculpture. The
- D) sculpture, as the

17

Publisher Mary Katharine Goddard (1738-1816) played a crucial role in the United States' fight for independence. Her newspaper, the Maryland Journal, emerged as a powerful voice for the patriot cause in the 1770s, advocating against British rule and _____ revolutionary ideals.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) had popularized
- B) popularizes
- C) would popularize
- D) popularizing

18

Paris Opera Ballet dancer Marie Camargo introduced a seemingly simple innovation to the ballet world of the 1730s by removing the heels from her dance shoes, creating the precursor to today's soft ballet slipper. This change was more than an aesthetic choice, _____ redesigned footwear enabled

her execution of leaps and other more athletic movements that had been impossible in the traditional heeled ballet shoes.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) however her
- B) however. Her
- C) however and her
- D) however, her

19

In 1909, children in the US were introduced to Billy Possum, a stuffed toy opossum named after the newly elected president, and William Taft, _____ by the entrepreneur Susie Wright Allgood. Billy Possum was promoted as a replacement for the popular teddy bear, which was named for the previous president, Theodore Roosevelt.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) invent
- B) is invented
- C) invents
- D) invented

20

Drought avoidance mechanisms—morphological adaptations that plants undergo when exposed to drought conditions—take a variety of forms. The soybean plant elongates its primary root to increase water _____ in the rosemary plant, drought triggers a decrease in the plant's stalk length that limits water loss.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) uptake, for instance, while
- B) uptake; for instance,
- C) uptake. For instance, while
- D) uptake, for instance,

21

With vast petroleum reserves, Saudi Arabia is the world's leading producer of crude oil, but the country's capacity to refine what it produces is surprisingly _____ country that imports nearly all its crude oil, Saudi Arabia refines only slightly more petroleum.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) modest --- compared to peer nation South Korea --- a
- B) modest. Compared to peer nation South Korea, a
- C) modest, compared to peer nation South Korea. A
- D) modest compared to peer nation South Korea, a

22

In biology there is no single agreed-upon concept of what defines a species; ____ numerous competing species concepts are used within the field, a dilemma known as the species problem. The biological species concept, described in 1942 by Mayr, defines species in accordance with reproductive isolation, while the phylogenetic species concept, delineated by Queiroz and Donoghue in 1988, organizes species according to monophyletic groups.

Which choice completes the text with the most logical transition?

- A) in addition,
- B) for instance,
- C) as a result,
- D) however,

23

Massimo Coli and colleagues used ground-penetrating radar (GPR) to map moisture infiltration within the walls of the Basilica di Santa Croce in Florence, Italy. The contrast between wet and dry materials produces vivid radar echoes that delineate leak pathways, yet the presence of water-saturated zones also hampers the efficacy of the technique. _____ even as GPR makes such areas conspicuous, water within masonry absorbs and scatters radar energy, often obscuring defects in layers beyond the saturated region.

Which choice completes the text with the most logical transition?

- A) Indeed,
- B) Thus,
- C) Likewise,
- D) Nevertheless,

24

In a 1998 study by Smith et al., the researchers used microhistological (fecal) analysis to determine the ratio of three plant subtypes (graminoids, forbs, and browse) within the diets of North American ungulates. The researchers did not perform this analysis on all such ungulates, ____ they focused exclusively on pronghorns and wild horses in New Mexico.

Which choice completes the text with the most logical transition?

- A) regardless;
- B) for instance;
- C) however;
- D) in other words;

25

While researching a topic, a student has taken the following notes:

- Carrots are vegetables that contain ascorbic acid, an essential nutrient for humans.
- There is 6 milligrams (mg) of ascorbic acid in every 100 grams (g) of carrot.
- Cabbage is a vegetable that contains ascorbic acid.

- There is 30 mg of ascorbic acid in every 100g of cabbage.
- Humans cannot make ascorbic acid in their bodies, so they must get it from foods, including fruits and vegetables.
- Ascorbic acid is also known as vitamin C.

The student wants to emphasize a similarity between carrots and cabbage. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Cabbage is a vegetable that contains ascorbic acid; in fact, there is 30 mg of it in every 100g of cabbage.
- B) Just like fruits, vegetables are a food that humans can eat to get ascorbic acid, also known as vitamin C.
- C) One thing that carrots and cabbage have in common is that they both contain vitamin C.
- D) Humans cannot make vitamin C in their bodies, so they must get this essential nutrient from vegetables, such as carrots.

26

While researching a topic, a student has taken the following notes:

- The Shinkansen, commonly known as the bullet train, is a network of high-speed railways in Japan.
- 1960s: The trains' high speeds created intense pressure waves that produced sonic booms when the trains exited tunnels.
- 1980s: Engineer Eiji Nakatsu noted that the kingfisher bird could dive into water at high speeds while barely creating a splash.
- The kingfisher's beak gradually widens from tip to head, allowing the bird to transition smoothly between air and water.
- Nakatsu and the other engineers redesigned the train nose based on the kingfisher's beak structure to improve the train's transition between tunnel and open-air environments.
- The redesign reduced pressure buildup, eliminating sonic booms.

The student wants to explain how the kingfisher's beak structure was used to solve the bullet train's sonic boom problem. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) The shape of the kingfisher's beak inspired engineers to reshape the train nose into one that gradually widens, allowing the train to move smoothly between air and water.
- B) Nakatsu's observation of kingfishers diving into water showed how the bird's beak structure could help the train eliminate sonic booms.
- C) Nakatsu discovered that the kingfisher's beak structure eliminated sonic booms, and the redesign improved the train's transition between tunnel and open-air environments.
- D) By redesigning the train's nose to imitate the kingfisher's gradually widening beak shape, engineers improved the train's transition between tunnels and open air, eliminating sonic booms.

While researching a topic, a student has taken the following notes:

- The French Republican calendar replaced the Gregorian calendar in France from 1793 to 1805.
- Both calendars are divided into twelve months.
- The timelines for the months don't match.
- The seventh month of the French Republican calendar, Germinal, begins during March, the third month of the Gregorian calendar.
- The twelfth month of the French Republican calendar, Fructidor, begins during August, the eighth month of the Gregorian calendar.

The student wants to emphasize a difference between the French Republican and Gregorian calendars. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) The French Republican calendar, which replaced the Gregorian calendar in France from 1793 to 1805, has twelve months, including the months of Germinal and Fructidor.
- B) The timelines for the months of the two calendars don't match; for example, the seventh month of the French Republican calendar begins during the third month of the Gregorian calendar.
- C) Used in France from 1793 to 1805, the French Republican calendar includes the months of Germinal and Fructidor (the seventh and twelfth months of the year).
- D) The third and eighth months of the Gregorian calendar are March and August, respectively.

Reading and Writing Module 1 Answers"

1. C
2. C
3. A
4. A
5. B
6. A
7. C
8. C
9. B
10. A
11. C
12. D
13. B
14. C
15. C
16. A

- 17. A
- 18. B
- 19. C
- 20. A
- 21. B
- 22. A
- 23. A
- 24. A
- 25. A
- 26. B
- 27. B

Reading and Writing Module 2 Answers"

- 1. D
- 2. C
- 3. D
- 4. A
- 5. C
- 6. D
- 7. B
- 8. B
- 9. A
- 10. B
- 11. C
- 12. C
- 13. D
- 14. B
- 15. B
- 16. A
- 17. D
- 18. B
- 19. D
- 20. A
- 21. B
- 22. C
- 23. A
- 24. D
- 25. C
- 26. D
- 27. B

Section 2, Module 1: Math

1 SAT Math

A right square prism has a height of 10 units. The volume of the prism is 160 cubic units. What is the length, in units, of an edge of the base?

2 SAT Math

If $9x+3=34$, what is the value of $2(9x+3)$?

- A. 62
- B. 68
- C. 74
- D. 102

3 SAT Math

The function f is defined by $f(x) = 2x^2 - 3x$. What is the value of $f(6)$?

4 SAT Math

In the xy -plane, what is the radius of the circle defined by $(x + 2)^2 + (y + 6)^2 = 289$?

5 SAT Math

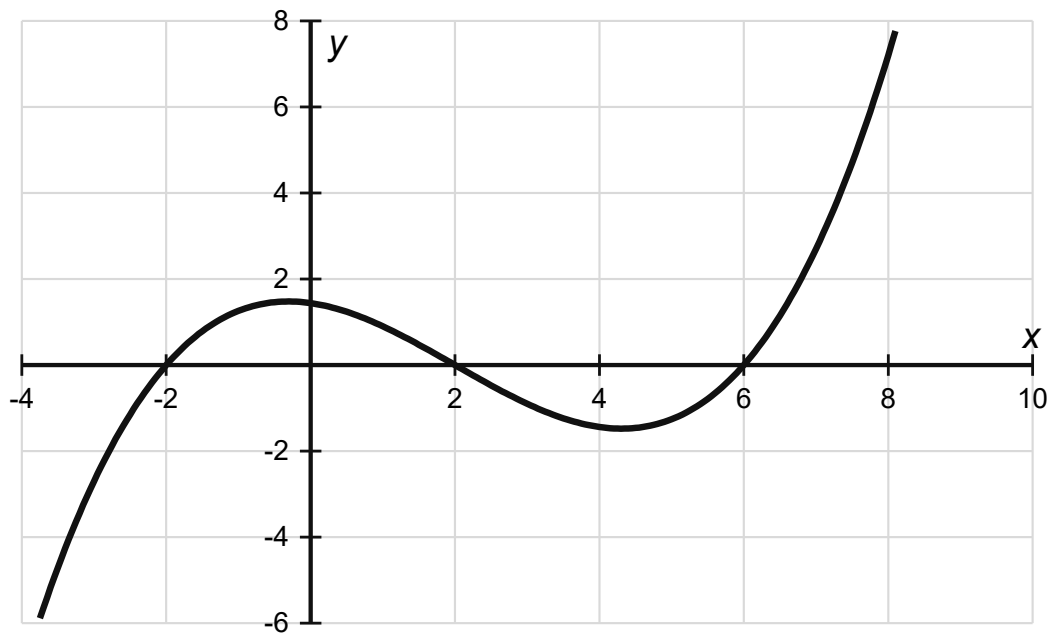
At what point (x, y) do the graphs of the equations $y = 9$ and $y = -2x + 7$ intersect in the xy -plane?

- A. $(9, -2)$
- B. $(-1, 9)$
- C. $(9, 9)$
- D. $(9, 7)$

6 SAT Math

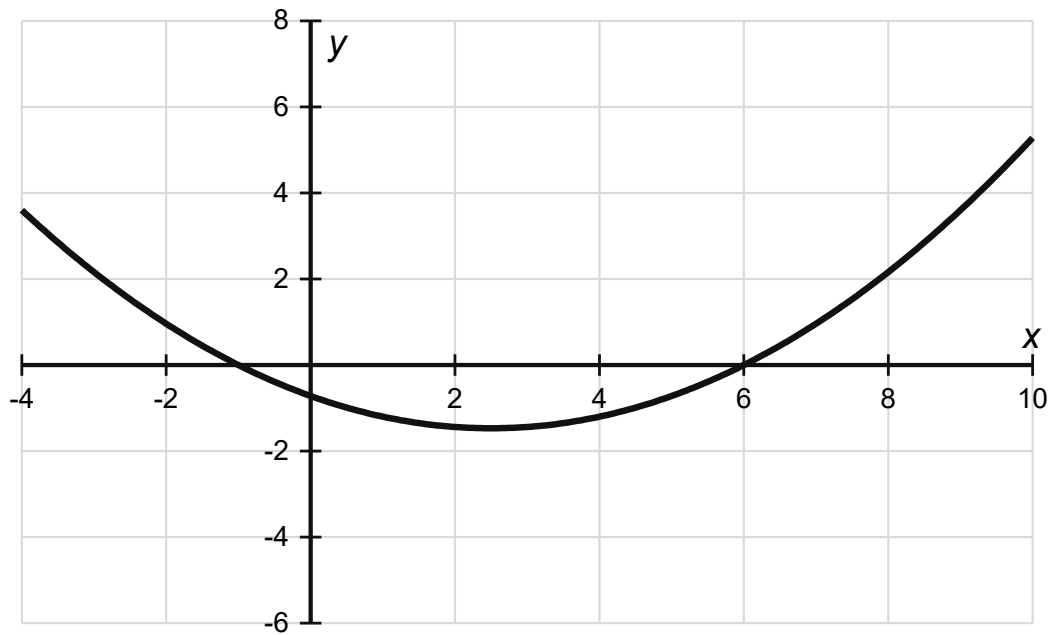
$(8y + 6) - (5y + 1)$ The given expression is equivalent to $by + c$, where b and c are constants. What is the value of b ?

The graph of the polynomial function f is shown, where $y = f(x)$. What is the value of $f(6)$?



- A. 6
- B. 2
- C. 0
- D. -3

The graph of $y = f(x)$ is shown in the xy -plane. For what value of x does $f(x) = 0$?



- A. 6
- B. 1
- C. 0
- D. -6

9 SAT Math

$g = 12 - (x)/(23)$ The equation shown gives the estimated amount of gas g , in gallons, that remains in the gas tank of a car after being driven x miles, where $0 \leq x \leq 276$. What is the estimated amount of gas, in gallons, that remains in the gas tank of the car when $x = 230$?

- A. 0
- B. 2
- C. 11
- D. 12

10 SAT Math

In the linear function f , $f(0) = 6$ and $f(7) = 6$. Which equation defines f ?

- A. $f(x) = 0$
- B. $f(x) = 7$
- C. $f(x) = 6$
- D. $f(x) = x + 6$

11 SAT Math

Of the 180 soccer players at a soccer camp, 72 are midfielders. What percentage of the soccer players at the camp are midfielders?

- A. 72%
- B. 40%
- C. 7.2%
- D. 0.4%

12 SAT Math

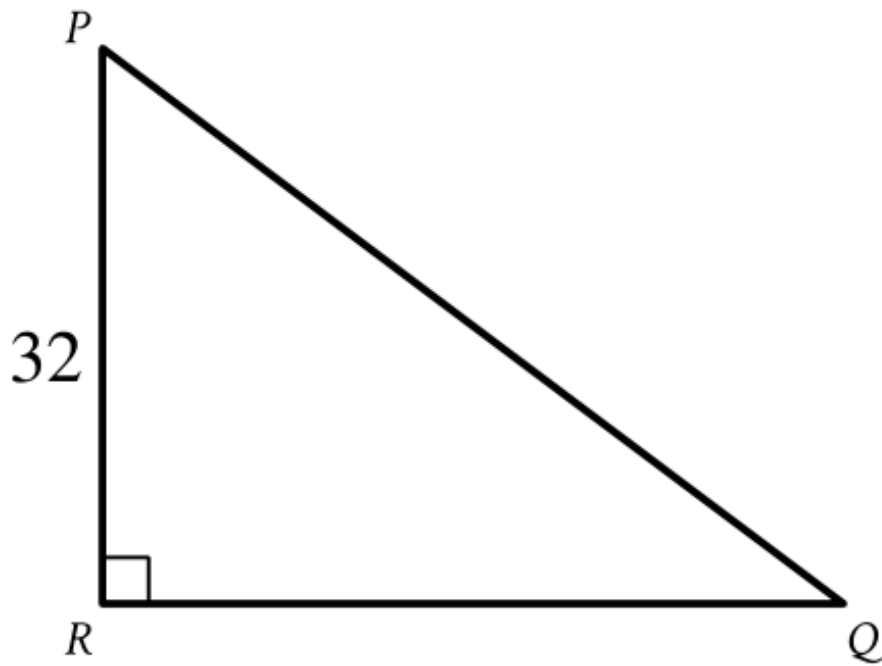
Which expression is equivalent to $7x^3 + 4x - 5 - 9x^2 + 6x + 3$?

- A. $7x^3 - 9x^2 + 10x - 2$
- B. $7x^3 - 9x^2 - 2x - 8$
- C. $-2x^3 + 2x - 8$
- D. $-2x^3 + 10x - 2$

13 SAT Math

The side lengths of right triangle RST are $RS = 280$, $ST = 96$, and $TR = 296$. Triangle RST is similar to triangle UVW, where S corresponds to V and T corresponds to W. What is the value of $\tan W$?

- A. $12/37$
- B. $12/35$
- C. $35/37$
- D. $35/12$

14 SAT Math

Note: Figure not drawn to scale.

In triangle PQR, the sine of $\angle Q$ is $\frac{1}{2}$. What is the length of segment PQ?

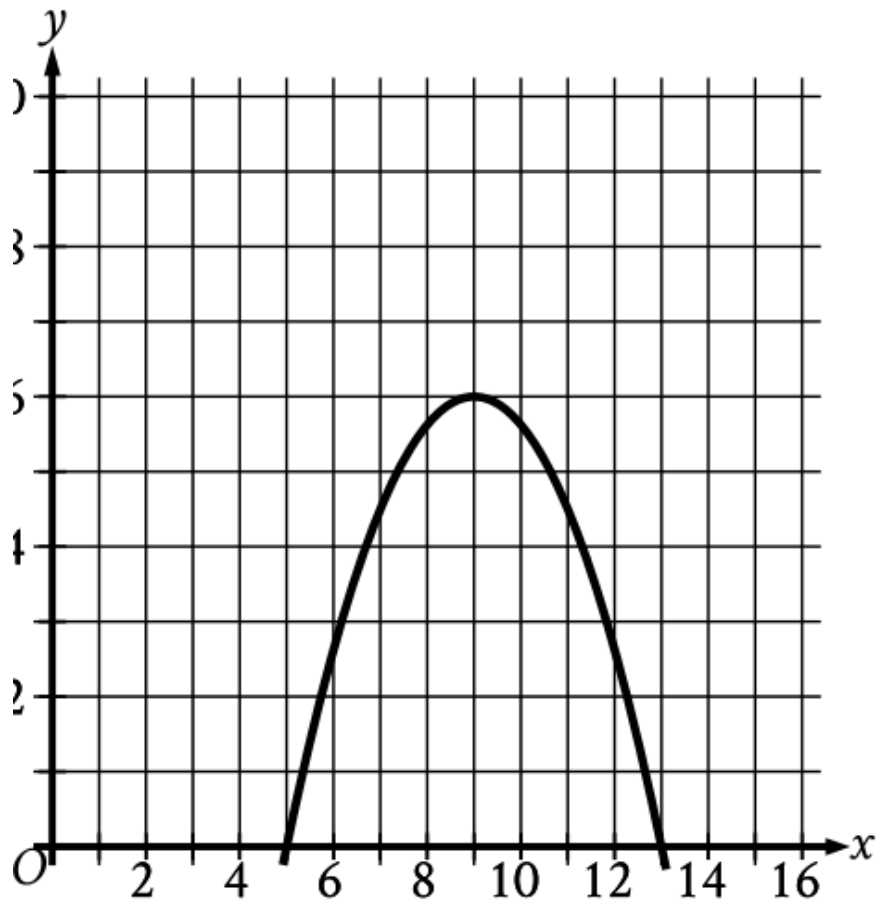
15 SAT Math

In the xy -plane, line k passes through the points $(6, 5)$ and $(7, 11)$ and is defined by the equation $y = mx + b$. What is the value of b ?

16 SAT Math

Maya is enlarging the area of a photograph that measures 7 centimeters by 5 centimeters. The resulting area, in square centimeters, of the photograph is given by the function $S(x) = (7x)(5x)$, where x represents the scale factor used to increase both the length and width of the photograph. Which statement is the best interpretation of $S(3) = 315$?

- A.** When the scale factor used to enlarge the photograph is 3, the resulting area is 7 square centimeters.
- B.** When the scale factor used to enlarge the photograph is 15, the resulting area is 35 square centimeters.
- C.** When the scale factor used to enlarge the photograph is 3, the resulting area is 315 square centimeters.
- D.** When the scale factor used to enlarge the photograph is 315, the resulting area is 3 square centimeters.



shown is $y = -\frac{3}{8}x - p^2 + 6$, where p is an integer constant

An equation of the graph shown is $y = -(3/8)(x - p)^2 + 6$, where p is an integer constant.
What is the value of p ?

18 SAT Math

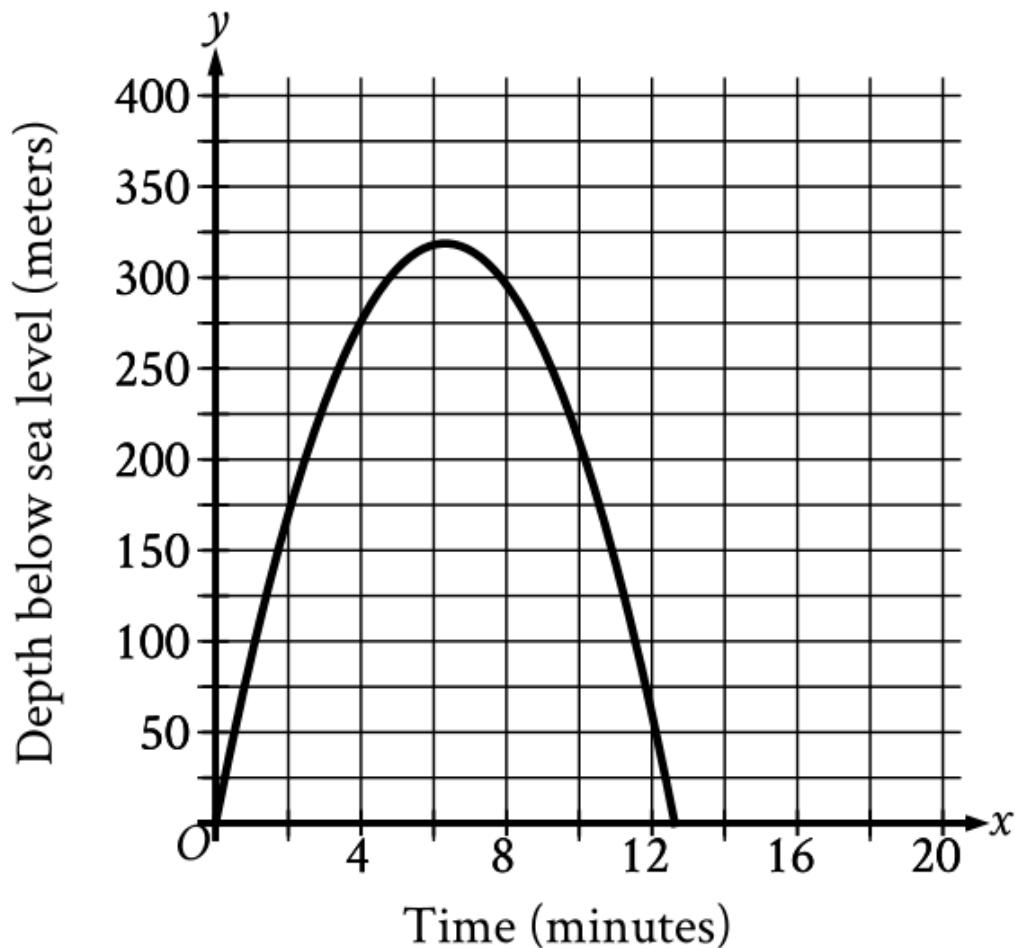
A bakery buys bags for \$0.06 each and boxes for \$0.38 each. The total cost of x bags and y boxes is \$524.72. Which equation represents this situation?

A. $0.38x - 0.06y = 524.72$

B. $0.38x + 0.06y = 524.72$

C. $0.06x + 0.38y = 524.72$

D. $0.06x - 0.38y = 524.72$



the predicted depth below sea level y , in meters, of a certain seal dives into the water. Based on the graph, which of the f

The graph shows the predicted depth below sea level y , in meters, of a certain seal x minutes after the seal dives into the water. Based on the graph, which of the following is closest to the predicted depth below sea level, in meters, of the seal 2 minutes after the seal dives into the water?

- A. 0
- B. 170
- C. 276

20 SAT Math

Turkey	a
Ham	b
Veggie	c
Total	156

A local deli sells three types of sandwiches: turkey, ham, and veggie. The table shows the number of sandwiches of each type that the deli has available to sell, where a , b , and c are distinct positive integer constants. If a sandwich is selected at random, which expression represents the probability of selecting a sandwich that is not ham?

- A. $\frac{a + c}{156}$
- B. $\frac{a + b + c}{156}$
- C. $\frac{b + c}{156}$
- D. $\frac{a + b}{156}$

21 SAT Math

Circle K has a radius of 4 millimeters (mm). Circle L has an area of 81π mm². What is the total area, in mm², of circles K and L?

- A. 26π
- B. 97π
- C. 13π
- D. 52π

22 SAT Math

Each side of equilateral triangle R is multiplied by a scale factor of k to create equilateral triangle U. The length of each side of triangle U is greater than the length of each side of triangle R. Which of the following could be the value of k ?

A. $\frac{31}{32}$

B. 1

C. $\frac{32}{31}$

D. 0

Section 2, Module 2: Math

1 SAT Math

A triathlon is a multisport race consisting of three consecutive events. A triathlete plans to complete a race at an average speed of 31,152 yards per hour. What is this average speed, in miles per hour? (1 mile = 1,760 yards)

2 SAT Math

If $4(8 - 3x) + 5 \geq 3(8 - 3x) + 19$, which of the following could be the value of $8 - 3x$?

- A. -2
- B. 2
- C. -14
- D. 14

3 SAT Math

0.0036 is 0.6% of a , and 108 is 450% of b , where a and b are positive numbers. The value of b is what percent of the value of a ?

- A. 40%
- B. 4,000%
- C. 144%
- D. 1,440%

4 SAT Math

A scientist studied the effects of mixing two chemicals, chemical A and chemical B, by gradually adding more of each chemical to a mixture of the two chemicals. At the start of the study, the mass of chemical A in the mixture was equal to the mass of chemical B in the mixture. From the start of the study to the end of the study, the mass of chemical A in the mixture increased by 2,400% and the mass of chemical B in the mixture increased by 280%. At the end of the study, approximately how many times greater was the mass of chemical A in the mixture than the mass of chemical B in the mixture?

- A. 21.20
- B. 6.58
- C. 26.80
- D. 8.57

5 SAT Math

The table shows three values of x and their corresponding values of $g(x)$, where $g(x) = f(x)/(x + 2)$ and f is a linear function.

x	$g(x)$
-1	36
0	20
2	12

What is the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane?

6 SAT Math

Which expression is equivalent to $5/(x - 8) + 5/(x - 8)^2$, where $x > 8$?

- A. $(5x - 45)/((x - 8)^2)$
- B. $(5x - 35)/((x - 8)^2)$
- C. $(25)/((x - 8)^2)$
- D. $(10)/((x - 8) + (x - 8)^2)$

7**SAT Math**

An object's speed is increasing at a rate of 3.6 meters per second squared. What is this rate, in miles per minute squared, rounded to the nearest tenth? (Use 1 mile = 1,609 meters.)

- A. 0.1
- B. 8.1
- C. 96.5
- D. 446.9

8**SAT Math**

$$tv - 49 + t = w$$

The given equation relates the positive numbers w , t , and v . Which equation correctly expresses v in terms of t and w ?

- A. $v = \frac{w - t}{t} + 48$
- B. $v = \frac{w}{t} - 49$
- C. $v = \frac{w}{t} + 48$
- D. $v = w - t - 49$

9**SAT Math**

The height of a right circular cylinder is 47 inches, and the circumference of its base is 376 inches. Which expression represents the total surface area, in square inches, of the cylinder?

A. $47(376) + \pi\left(\frac{188}{\pi}\right)^2$

B. $\pi\left(\frac{188}{\pi}\right)^2(47)$

C. $47(376) + 2\pi\left(\frac{188}{\pi}\right)^2$

D. $47(376)$

10**SAT Math**

In 2013, Chloe earned 11% more than in 2012, and in 2014 Chloe earned 6% more than in 2013. If Chloe earned y times as much in 2012 as in 2014, which of the following is closest to the value of y ?

A. 0.5455

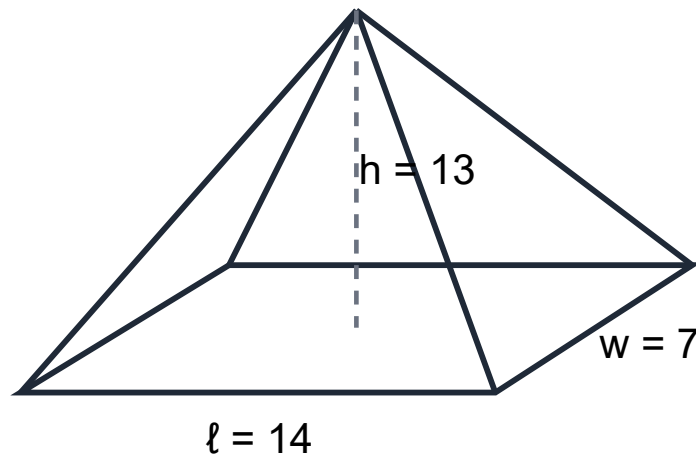
B. 0.6600

C. 0.8499

D. 1.1766

11 SAT Math

The figure shown is a right rectangular pyramid, where $\ell = 14$ units, $w = 7$ units, and $h = 13$ units. What is the surface area, in square units, of the pyramid?



Note: Figure not drawn to scale.

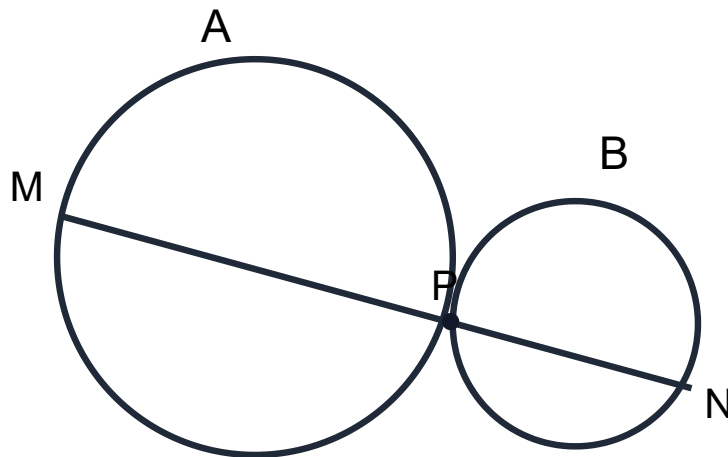
12 SAT Math

$3rx + 5 = 12x + 9$ In the given equation, r is a constant. The equation has no solution. What is the value of r ?

- A. 6
- B. 4
- C. 18
- D. 7

13 SAT Math

Spheres A and B have a point of tangency at point P, where segment MP is a diameter of sphere A, segment NP is a diameter of sphere B, and point P lies on segment MN. The radius of sphere A is 13 times the radius of sphere B, and the volume of sphere B is 288π cubic inches. What is the length, in inches, of segment MN?



Note: Figure not drawn to scale.

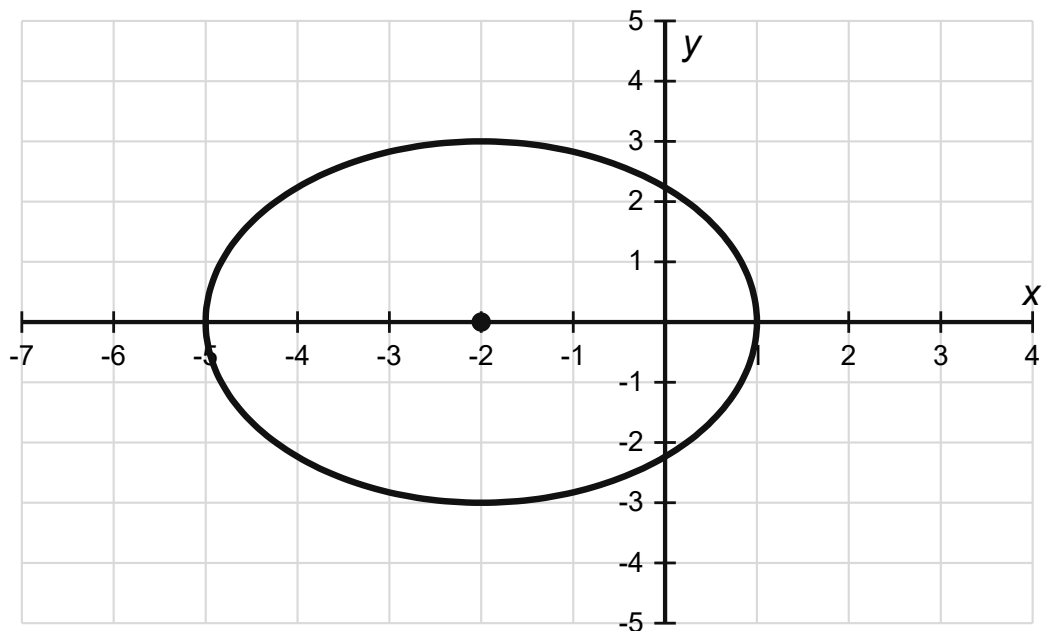
14 SAT Math

The function f is defined by $f(x) = ax^2 + bx + c$, where a , b , and c are constants. The graph of $y = f(x)$ in the xy -plane passes through the points $(9, 0)$ and $(-2, 0)$. If a is an integer greater than 1, which of the following could be the value of $a + b$?

- A. 8
- B. 7
- C. -6
- D. -12

15 SAT Math

Circle A (shown) is defined by the equation $(x + 2)^2 + y^2 = 9$. Circle B (not shown) is the result of shifting circle A down 4 units and increasing the radius so that the radius of circle B is 5 times the radius of circle A. Which equation defines circle B?



- A. $(x + 2)^2 + (y + 4)^2 = 25(9)$
- B. $5(x + 2)^2 + 5(y + 4)^2 = 9$
- C. $(x + 2)^2 + (y - 4)^2 = 25(9)$
- D. $5(x + 2)^2 + 5(y - 4)^2 = 9$

16 SAT Math

$x + 33 = \sqrt[3]{(x + 45)} - 12$ The solutions to the given equation are s and t , where $s < t$. What is the value of s ?

17 SAT Math

The function p is defined by $p(x) = a((x + 6)^2 - b)((x + 6)^2 - c)$, where a , b , and c are constants. In the xy -plane, the graph of $y = p(x)$ passes through the points $(-7, 42)$ and $(0, 72)$. What is the value of $p(-12) + p(-5)$?

18 SAT Math

For an online trivia game, 300 points are awarded for a correct answer if a question is answered in less than 4 seconds from the time the question is asked. Each 4 seconds after the question is asked, the number of points awarded for a correct answer decreases by 15% of the number of points awarded for a correct answer in the previous 4 seconds. Which function gives the number of points awarded for a correct answer x seconds from the time the question is asked, where x is a multiple of 4?

- A. $f(x) = 300(0.15)^{x/4}$
- B. $f(x) = 300(0.85)^{4x}$
- C. $f(x) = 300(0.85)^{x/4}$
- D. $f(x) = 300(0.15)^{4x}$

19 SAT Math

Which expression is equivalent to $(x^2 - 19x + 70)/(x^2 - 25)$, where $x > 5$?

- A. $(x + 14)/(x - 5)$
- B. $(x - 14)/(x + 5)$
- C. $(x - 14)/(x - 5)$
- D. $(x + 14)/(x + 5)$

20 SAT Math

I. $g(x) = a(2.7)^x + k$

II. $h(x) = m(2.7)^x + (2.7)^b$

The function f is defined by $f(x) = a(2.7)^x + (2.7)^b$, where a and b are integer constants and $0 < a < b$. The functions g and h are equivalent to function f , where k and m are constants. Which of the following equations displays the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane as a constant or coefficient?

- A. II only
- B. Neither I nor II
- C. I and II
- D. I only

21 SAT Math

A researcher surveyed a random sample of 1,243 people from a certain state to estimate the percentage of people who support the governor's recent approval of a bill. From the survey, the researcher estimated that 69% of people from the state support the governor's approval of the bill, with an associated margin of error of 2.57%. The researcher repeated the survey with a random sample of people from the state that was double the original sample size. Assuming the margins of error were calculated in the same way, which of the following is true about the margin of error associated with the estimate from the larger sample?

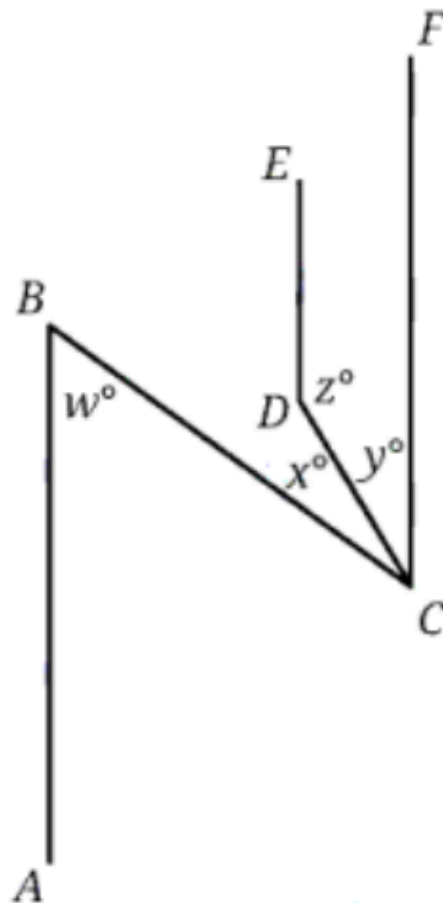
- A. The margin of error is between 2.57% and 3.57%.
- B. The margin of error is greater than 5.57%.
- C. The margin of error is between 4.57% and 5.57%.
- D. The margin of error is less than 2.57%.

22 SAT Math

In the figure, AB , DE , and CF are parallel. If $202^\circ < x + y + z < 215^\circ$, which of the following could be a value of $w - y$?

- I. 23°
- II. 35°

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II



Note: Figure not drawn to scale.

1	4	1	17.7
2	B	2	D
3	54	3	B
4	17	4	B
5	B	5	40
6	3	6	B
7	C	7	B
8	A	8	C
9	B	9	C
10	C	10	C
11	B	11	$98+35\sqrt{29}+7\sqrt{218}$
12	A	12	B
13	D	13	168
14	64	14	D
15	-31	15	A
16	C	16	-45
17	9	17	114
18	C	18	C
19	B	19	B
20	A	20	B
21	B	21	D
22	C	22	A